



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI RAM FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

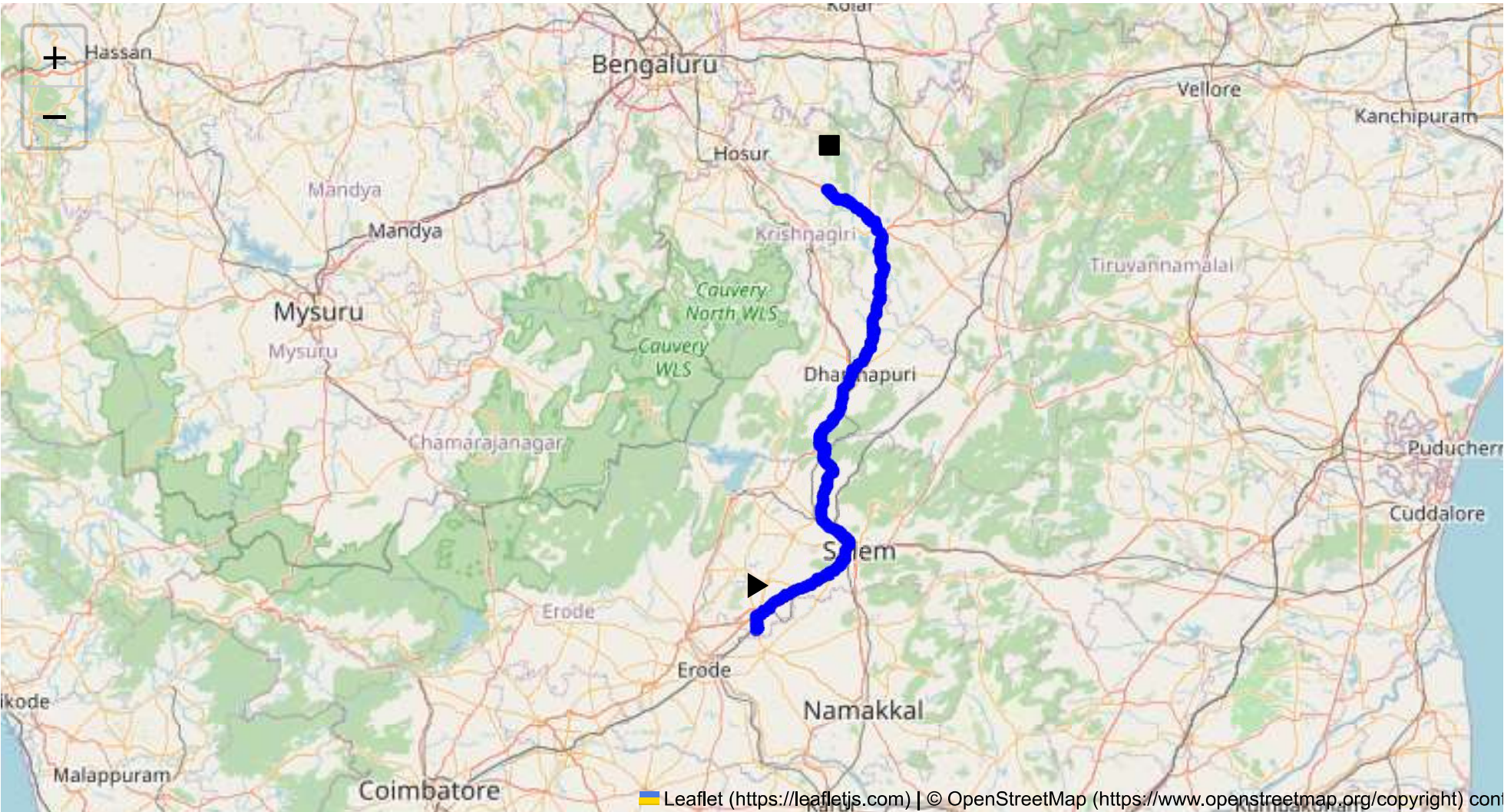
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 173.36 km
Estimated Duration: 3.1 hours
Adjusted Duration (Heavy Vehicle): 3.9 hours
Start: (11.4381, 77.8734)
End: (12.636431, 78.069847)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to Chinnar covers approximately 173.36 kilometers and typically requires around 3.09 hours for heavy vehicles. The path primarily follows well-established highways and arterial roads that connect these regions, incorporating stretches of NH44 and AH47, which are part of the Indian highway system. Key points along the route include cities such as Salem and Dharmapuri.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu experiences a tropical climate. During monsoon seasons (June to September and October to December), heavy rainfall can lead to waterlogging and reduced visibility, which may pose significant hazards. Summer months can be extremely hot, potentially leading to asphalt melting and tire blowouts.

3. Traffic Patterns and Congestion: Traffic congestion can be observed near urban areas such as Salem, particularly during morning and evening hours (8 AM - 10 AM and 6 PM - 8 PM). Highway stretches are generally less congested, but construction or maintenance work can periodically contribute to slowdowns.

4. Assessment of Road Quality and Infrastructure: The route comprises generally well-maintained highways, especially the sections on NH44 and AH47. While the major highways are in good condition, some connecting roads might be narrower with sporadic potholes, especially after monsoons.

5. Suggestions for Alternative Routes for Emergencies: In case of emergencies or roadblocks, alternative routes can involve regional roads connecting via smaller towns like Pochampalli or utilizing minor state highways, though they may add travel time and involve lower-quality infrastructure. It's advisable to have GPS navigation updated with real-time traffic data.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Tamil Nadu, like other Indian states, follows central government guidelines for the transport of hazardous materials. This includes strict adherence to the Motor Vehicles Act concerning vehicle fitness, speed limitations, restricted timings for travel in dense city areas, and the necessity for appropriate placards and documentation.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials: While precise incident data for this specific route is limited, highways in Tamil Nadu occasionally witness accidents involving heavy vehicles due to factors like over-speeding or mechanical failure. It's important to follow stringent maintenance checks and speed regulations.

8. Environmental Considerations and Sensitive Areas: Environmental sensitivity is notably high around regions with significant agricultural activity and near water bodies. Drivers should be aware of and adhere to environmental protection signs, especially in areas near protected lands or forests adjacent to highways.

9. Analysis of Communication Coverage: Mobile network coverage is robust along highways with occasional dead zones, especially in rural stretches or hilly terrains. It's advisable to have emergency numbers saved offline and a physical map as a backup.

10. Estimated Emergency Response Times: In urban zones like Salem, emergency response times might be relatively quick, within 15-30 minutes. In rural segments, response times can extend to over an hour due to less accessibility and the distance to emergency facilities.

12. Overall Summary of Risk Assessment: The route from Sangagiri to Chinnar involves moderate risks primarily due to weather conditions and urban congestion. Monitoring weather forecasts, adhering to regulations, and maintaining vehicle conditions can mitigate most hazards. Familiarity with alternate routes and communication preparedness can enhance safety during unforeseen events. Ensuring cargo security and compliance with hazardous material transport regulations further aids in minimizing risks.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45028, 77.87202	15 KM/Hr
4	Turn	Medium	11.46400, 77.87323	30 KM/Hr
5	Turn	Medium	11.46765, 77.87037	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47986, 77.88397	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.53501, 77.95707	30 KM/Hr
11	Turn	Medium	11.53601, 77.95856	30 KM/Hr
12	Turn	Medium	11.62403, 78.10893	30 KM/Hr
13	Turn	Medium	11.62917, 78.11648	30 KM/Hr
14	Turn	High	11.64180, 78.11808	15 KM/Hr
15	Turn	High	11.64325, 78.11993	15 KM/Hr
16	Turn	Medium	11.67865, 78.12803	30 KM/Hr
17	Turn	Medium	11.68201, 78.12639	30 KM/Hr
18	Turn	Medium	11.69519, 78.10775	30 KM/Hr
19	Turn	Medium	11.73958, 78.05105	30 KM/Hr
20	Turn	Medium	11.75835, 78.05016	30 KM/Hr
21	Turn	Medium	11.86847, 78.08367	30 KM/Hr
22	Turn	Medium	11.87942, 78.08197	30 KM/Hr
23	Turn	Medium	11.92585, 78.06507	30 KM/Hr
24	Turn	Medium	11.93319, 78.06419	30 KM/Hr
25	Turn	Medium	11.93557, 78.06103	30 KM/Hr
26	Turn	Medium	11.93704, 78.05402	30 KM/Hr
27	Turn	High	11.95818, 78.04978	15 KM/Hr
28	Turn	Medium	11.95988, 78.04948	30 KM/Hr
29	Turn	Medium	11.96304, 78.05136	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
30	Turn	High	11.96399, 78.05135	15 KM/Hr
31	Turn	High	11.96580, 78.04923	15 KM/Hr
32	Turn	Medium	11.96675, 78.04939	30 KM/Hr
33	Turn	Medium	11.96791, 78.05070	30 KM/Hr
34	Turn	Medium	12.16983, 78.16769	30 KM/Hr
35	Turn	Medium	12.19881, 78.18157	30 KM/Hr
36	Turn	Medium	12.22459, 78.19136	30 KM/Hr
37	Turn	Medium	12.24731, 78.19601	30 KM/Hr
38	Turn	Medium	12.25060, 78.19977	30 KM/Hr
39	Turn	Medium	12.25329, 78.20039	30 KM/Hr
40	Turn	Medium	12.29110, 78.19909	30 KM/Hr
41	Turn	Medium	12.39904, 78.21764	30 KM/Hr
42	Turn	Medium	12.40156, 78.21845	30 KM/Hr
43	Turn	Medium	12.40589, 78.22275	30 KM/Hr
44	Turn	Medium	12.45154, 78.21792	30 KM/Hr
45	Turn	Medium	12.50553, 78.22318	30 KM/Hr
46	Turn	Medium	12.59749, 78.13593	30 KM/Hr
47	Turn	Medium	12.63696, 78.07061	30 KM/Hr
48	Turn	High	12.63717, 78.06938	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
1	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
2	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
3	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
4	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
6	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
8	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
9	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
10	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
11	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
13	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
18	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
19	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
20	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
22	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
23	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
24	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
25	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
26	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
27	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium
30	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
31	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
32	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
33	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
34	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
36	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
41	hospital	TCR Multi Speciality Hospital	12.5088183, 78.2237657	30 km/h	Medium
43	hospital	KVS Hospital	12.5277857, 78.2055049	30 km/h	Medium
44	hospital	Ga Heart Hospital	12.5287708, 78.2055336	30 km/h	Medium
45	clinic	Psychiatric Centre	12.529476, 78.205064	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
12	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
14	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
15	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
16	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
17	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
21	college	Sona College	11.6768, 78.1257052	30 km/h	Medium
28	university	Periyar University	11.7186758, 78.0771089	30 km/h	Medium
29	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
35	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
37	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
38	school	Nalanda Metric Higher Secondary School	12.488407, 78.21856	30 km/h	Medium
39	school	Nalanda International Public School	12.4994979, 78.2235094	30 km/h	Medium
40	school	Don Bosco School Ground	12.505954, 78.2222003	30 km/h	Medium
42	school	Krishland Higher Secondary School	12.5181566, 78.2178556	30 km/h	Medium
46	school	Sri Saraswathi Vidyalaya	12.5620799, 78.183049	30 km/h	Medium
47	college	Government Medical College	12.582332, 78.1496363	30 km/h	Medium
48	school	Crescent Matriculation Higher Seconddary School	12.6003724, 78.132787	30 km/h	Medium
49	college	Jeeva College of Nursing	12.6067058, 78.1191944	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.47986, 77.88397



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53501, 77.95707



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53601, 77.95856



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62403, 78.10893



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62917, 78.11648



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64180, 78.11808



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64325, 78.11993



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67865, 78.12803



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68201, 78.12639



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69519, 78.10775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73958, 78.05105



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75835, 78.05016



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86847, 78.08367



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Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



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Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



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Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93319, 78.06419



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93557, 78.06103



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



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Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



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Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



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Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



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Speed Limit: 30 KM/Hr
Coordinates: 12.16983, 78.16769



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Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19881, 78.18157



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Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22459, 78.19136



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Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24731, 78.19601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25060, 78.19977



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25329, 78.20039



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.29110, 78.19909



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.39904, 78.21764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40156, 78.21845



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.40589, 78.22275



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.45154, 78.21792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.50553, 78.22318



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.59749, 78.13593



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.63696, 78.07061

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.63717, 78.06938